

Supplementary Information

A Judge Agent Closes the Reliability Gap in AI-Generated Scientific Simulation

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S1 Error Bound Proof within the Simulability Class $\mathfrak{S}$

We restate and prove the main formal guarantee, which applies to problems in the simulability class $\mathfrak{S}$ (Definition 1, main text).

Proposition 1 (Automated Bounded-Error Realizability within $\mathfrak{S}$). *Under the following assumptions:*

- (A1) $P \in \mathfrak{S}$ (the simulability class, Definition 1 in main text): P admits decomposition into a DAG over K computational primitives, each well-posed and stably discretizable, with bounded Lipschitz composition.
- (A2) Hadamard well-posedness: for each primitive p_i , the associated operator has finite Lipschitz constant $L_i < \infty$ (condition 2 of $\mathfrak{S}$).
- (A3) Lax–Richtmyer convergence: each primitive p_i admits a discretization with error $\varepsilon_i \leq C_i h_i^{q_i}$ for mesh parameter h_i and convergence order $q_i > 0$ (condition 3 of $\mathfrak{S}$).
- (A4) Bounded composition: the DAG Lipschitz constant $L_{\text{DAG}} = \max_i \prod_{j \in \text{desc}(i)} L_j < \infty$ (condition 4 of $\mathfrak{S}$).

For any $\varepsilon > 0$, the three-agent pipeline produces $\hat{x}$ satisfying $\|\hat{x} - x^*\|_X \leq \varepsilon$ in finite time $T_{\text{comp}} < \infty$.

Proof. The proof proceeds in four steps.

Step 1: Finite specification. By condition (1) of $\mathfrak{S}$, P admits a finite representation $\mathcal{S} = (\Omega, \mathcal{E}, \mathcal{B}, \mathcal{I}, \mathcal{O}, \varepsilon)$. The Plan Agent ($\mathcal{A}_P$) generates this from natural language; correctness is verified by the Judge Agent ($\mathcal{A}_J$) via $\mathfrak{S}$ -membership verification. If $\mathcal{A}_J$ rejects, the pipeline either redesigns (up to 3 iterations) or terminates with a formal rejection certificate indicating which $\mathfrak{S}$ condition is violated.

Step 2: DAG decomposition. By condition (1), the discretized system decomposes into a finite directed acyclic graph $\mathcal{G} = (V, E)$ where each node $v_i \in V$ corresponds to a primitive $p_{k(i)}$ from the 12-primitive basis. Let $D = |V|$ denote the number of nodes (DAG size) and let the graph have depth $d \leq D$.

For each node v_i , the associated primitive $p_{k(i)}$ maps inputs to outputs with Lipschitz constant L_i (assumption A2). The primitive’s discretization has error bounded by $\varepsilon_i \leq C_i h_i^{q_i}$ (assumption A3).

Step 3: Error propagation through the DAG. We bound the total error by induction on the DAG structure. For a single primitive, the error is $\varepsilon_1 \leq C_1 h_1^{q_1}$. For a chain of two primitives $p_2 \circ p_1$:

$$\begin{aligned} \|p_2(p_1(\hat{x})) - p_2(p_1(x^*))\| &\leq L_2 \|p_1(\hat{x}) - p_1(x^*)\| + \varepsilon_2 \\ &\leq L_2(L_1\varepsilon_0 + \varepsilon_1) + \varepsilon_2 \end{aligned} \tag{1}$$

where $\varepsilon_0 = 0$ (exact input to the first primitive).

By induction, for a general DAG with nodes $v_1, \dots, v_D$ in topological order:

$$\varepsilon_{\text{total}} = \|\hat{x} - x^*\| \leq \sum_{i=1}^D \ell_i \cdot \varepsilon_i \tag{2}$$

where $\ell_i = \prod_{j \in \text{desc}(i)} L_j$ is the effective amplification factor from node v_i to the output. By the bounded composition assumption (A4), each $\ell_i \leq L_{\text{DAG}} < \infty$. The crude bound gives:

$$\varepsilon_{\text{total}} \leq D \cdot L_{\text{DAG}} \cdot \max_i \varepsilon_i \tag{3}$$

Step 4: Resolution selection. To achieve $\varepsilon_{\text{total}} \leq \varepsilon$, set:

$$\varepsilon_i = \frac{\varepsilon}{D \cdot \ell_i} \quad (4)$$

This ensures $\sum_i \ell_i \varepsilon_i = \sum_i \varepsilon / D = \varepsilon$. By assumption (A3), each ε_i is achievable by choosing:

$$h_i \leq \left(\frac{\varepsilon}{D \cdot \ell_i \cdot C_i} \right)^{1/q_i} \quad (5)$$

Each primitive with mesh parameter h_i has finite computational cost (condition (1) of $\mathfrak{S}$). The total cost is:

$$T_{\text{comp}} = \sum_{i=1}^D T_i(h_i) < \infty \quad (6)$$

since each T_i is finite for fixed $h_i > 0$. The Execute Agent ($\mathcal{A}_E$) sets these resolution parameters via the a priori estimates and verifies the bound a posteriori. $\square$

Connection to the Judge Agent’s gates. The Judge Agent’s 5 pre-execution gates directly verify the assumptions of Proposition 1:

- **Gates 1 & 4** (dimensional consistency, problem classification) verify assumption (A1): the problem admits a correct DAG decomposition into known primitives.
- **Gates 2 & 3** (BC/IC compatibility, well-posedness) verify assumptions (A2) and (A3): each primitive is well-posed and admits a convergent discretization.
- **Gate 5** (cost estimation) verifies that the resolution parameters h_i required by Step 4 are computationally feasible given assumption (A4).

Remark. The mathematical content is entirely classical—the error propagation through composed operators with Lipschitz bounds is standard numerical analysis (see, e.g., Lax & Richtmyer, 1956). The contribution of our pipeline is: (a) $\mathcal{A}_P$ automatically decomposes the problem into the DAG; (b) $\mathcal{A}_J$ verifies $\mathfrak{S}$ -membership (assumptions A1–A4); (c) $\mathcal{A}_E$ computes the resolution parameters and verifies the bound. No step requires mathematical novelty.

Where the bound breaks down. At the *scientific event horizon* (boundary $\partial\mathfrak{S}$), assumption (A4) fails: $L_{\text{DAG}} \rightarrow \infty$, typically because a bifurcation parameter $\theta \rightarrow \theta_c$ causes $\ell_i \sim |\theta - \theta_c|^{-\alpha}$ for some node i . Near $\partial\mathfrak{S}$, the required resolution $h_i \rightarrow 0$ and the computational cost $T_{\text{comp}} \rightarrow \infty$, making the bound vacuous. The two residual failures (1.5%) in the main text both occur in this regime.

S2 The Scientific Event Horizon: Four Obstructions

The simulability class $\mathfrak{S}$ (Definition 1, main text) delineates problems amenable to automated bounded-error simulation. Outside $\mathfrak{S}$ —at and beyond the scientific event horizon—four fundamental obstructions prevent any computational system (human or automated) from guaranteeing bounded-error solutions.

Proposition 2 (Obstructions at the Scientific Event Horizon). *The complement $\mathfrak{U} = \mathfrak{S}^c$ contains problems exhibiting at least one of four obstructions:*

Obstruction 1: Logical undecidability (Gödel–Turing). Problems reducible to the halting problem or Hilbert’s tenth problem over $\mathbb{Z}$. Example: “Does program P halt on input x ?” No algorithm—and therefore no agent pipeline—can decide this for all (P, x) . These problems violate $\mathfrak{S}$ condition (1): no finite DAG decomposition exists.

Obstruction 2: Real uncomputability (Blum–Shub–Smale). Decision problems over $\mathbb{R}$ requiring infinite-precision predicates. Example: Mandelbrot set membership for general $c \in \mathbb{C}$ requires determining whether an orbit remains bounded, which is BSS-undecidable for boundary points. These problems violate $\mathfrak{S}$ condition (3): no convergent discretization exists at the decision boundary.

Obstruction 3: Quantum measurement irreducibility. Individual measurement outcomes governed by the Born rule. Only probability distributions—not individual realizations—are computable. Example: “Which detector clicks in a Stern–Gerlach experiment?” is inherently stochastic. Note: the *distribution* of outcomes is in $\mathfrak{S}$ (via Monte Carlo sampling). This obstruction violates $\mathfrak{S}$ condition (2): the individual-outcome “problem” is not well-posed (no unique solution exists).

Obstruction 4: Infinite-precision barriers. Solutions depending on non-computable real numbers such as Chaitin’s Ω (the halting probability of a universal Turing machine). No finite computation can approximate Ω to arbitrary precision. This violates $\mathfrak{S}$ condition (3): no convergent approximation scheme exists.

Whether these four obstructions are the *only* barriers to membership in $\mathfrak{S}$ remains an open question. We state this as a conjecture grounded in the current state of computability theory, not as a proven result.

Practical implications for the Judge Agent. The four obstructions are theoretical; in practice, the scientific event horizon is encountered through *near-violations* of $\mathfrak{S}$ conditions—particularly condition (4) (bounded Lipschitz composition) near bifurcation points. The Judge Agent’s current gates detect exact violations (e.g., clearly ill-posed problems, missing BCs) but cannot detect proximity to $\partial\mathfrak{S}$. The bifurcation-sensitive rejection heuristics proposed in the main text (parameter continuation, Lyapunov exponent estimation, ensemble perturbation) aim to extend the Judge’s detection capability to near-violations of condition (4).

S3 Six Additional Domain Validations

In addition to the three deeply validated domains in the main text (CT, seismic, combustion), we validated the pipeline on seven additional domains, all within $\mathfrak{S}$. The first uses real reported data (COVID-19); the remaining six use synthetic ground truth.

Table 1. Table S0. Seven additional domain validations, all within $\mathfrak{S}$. $\mathfrak{S}$ conditions verified by the Judge Agent for each.

#	Domain	Problem	Primitives	Target	Achieved	Data
1	Epidemiology	COVID-19 SEIR-D (8 counties)	N, E, O, B	10% peak	12.4%	R
2	Classical Mech.	Granular flow (2D, $N=5000$)	∂, N, E, K, B, G	10^{-3}	7.2×10^{-4}	S
3	Electromagnetics	Waveguide modes (sanity)	∂, L, F, B, G	10^{-8}	2.1×10^{-9}	S
4	Quantum Chem.	He ground state (CI)	$\partial, L, \Pi, B, G, O$	1 mHa	0.4 mHa	S
5	Fluid Dynamics	BFS turbulent ($Re=5100$)	$\partial, E, N, K, B, G, \Pi$	5% TKE	3.8%	R [†]
6	Structural Mech.	Topology-opt. bracket	∂, L, O, B, G	10^{-3}	6.1×10^{-4}	S
7	Optics	Fresnel diffraction (sanity)	∂, F, L, B, G	10^{-5}	1.6×10^{-6}	S

R = real reported data. S = synthetic ground truth. [†]Validated against JHU Turbulence Database DNS data.

COVID-19 epidemic dynamics (8 U.S. counties). SEIR-D compartmental model with county-specific contact rates, fitted to Johns Hopkins CSSE data (Dong et al., 2020) from March–August 2020. Eight counties: Fulton (GA), Cook (IL), Los Angeles (CA), Harris (TX), Maricopa (AZ), King (WA), Miami-Dade (FL), Wayne (MI). Training: first 60% of data; validation: remaining 40%. Framework peak timing error: $12.4 \pm 3.1\%$; cumulative case count RMSE: $18.7 \pm 5.2\%$; $R^2 = 0.82 \pm 0.09$ (leave-one-county-out cross-validation). Expert epidemiological model (county-specific, manually calibrated SEIR with non-pharmaceutical intervention timing): $8.9 \pm 2.4\%$. Per-county results in Table S7.

$\mathfrak{S}$ -membership note: The SEIR-D ODE system is well-posed (Lipschitz RHS) and admits convergent discretization (BDF/RK methods). The problem lies well within $\mathfrak{S}$; the limitation is the *model class*, not the numerical method. The framework correctly identifies and applies the appropriate model class but cannot overcome the model’s inherent ceiling. We exclude this domain from the main text because this model-class limitation makes it a case study rather than a powered validation. Quality audit: population conservation, compartment non-negativity, cumulative death monotonicity; 0/8 flagged. Setup-time efficiency: $\rho = 240\times$ (15 min vs. 2.5 days expert).

Granular flow. 5,000-particle Hertz–Mindlin DEM simulation. $\mathfrak{S}$ verification: the N-body contact dynamics is well-posed for finite particles with bounded forces; the velocity-Verlet integrator is stable under the adaptive CFL condition. The framework correctly identifies the need for adaptive time-stepping near collision events. $\rho = 560\times$.

Waveguide modes. Rectangular waveguide eigenvalue problem solved via Helmholtz equation discretization. A textbook-level sanity check ($\mathfrak{S}$ membership trivially verified) confirming the pipeline handles standard eigenvalue problems. Error: 2.1×10^{-9} relative to analytical TE/TM mode frequencies. $\rho = 480\times$.

Helium ground state. Configuration interaction (CI) calculation with STO-6G and cc-pVDZ basis sets. $\mathfrak{S}$ membership: the CI eigenvalue problem is a finite-dimensional linear algebra problem (well-posed, convergent, bounded). Error: 0.4 mHa vs. exact non-relativistic value, consistent with basis set limitation. $\rho = 720\times$.

Backward-facing step (BFS). Turbulent flow at $Re = 5100$ validated against JHU Turbulence Database DNS. The framework selects LES with Smagorinsky subgrid model on a $256 \times 128 \times 64$ grid. Note: this problem approaches the scientific event horizon—turbulent flows at higher Re would violate $\mathfrak{S}$ condition (4) as the Lipschitz constant grows. At $Re = 5100$, the LES formulation remains within $\mathfrak{S}$. TKE profiles match DNS within 3.8%. $\rho = 916\times$.

Topology optimization. SIMP method for a bracket under distributed load. The framework correctly implements penalization, sensitivity filtering, and optimality criteria update. $\mathfrak{S}$ membership verified via the well-posedness of the compliance minimization problem with volume constraint. $\rho = 288\times$.

Fresnel diffraction. Single-slit diffraction pattern computed via Fresnel–Kirchhoff integral. Sanity check; error 1.6×10^{-6} vs. analytical result. $\rho = 432\times$.

S4 Primitive Basis and Minimality

The 12 computational primitives define the operations from which all DAG decompositions within $\mathfrak{S}$ are constructed. We prove that each primitive is necessary by exhibiting a witness problem $P_k \in \mathfrak{S}$ that requires primitive p_k and cannot be solved using only the remaining 11.

Witness problems for each primitive:

1. **Differentiate** (∂): Poisson equation $-\nabla^2 u = f$ on $[0, 1]^2$. Requires spatial derivatives; no

Table 2. Table S1. 25 numerical method families decomposed over the 12-primitive basis. All 25 families have been verified to satisfy $\mathfrak{S}$ conditions (2)–(4) for standard problem instances.

Method Family	Primitive Decomposition
Finite Difference (FD)	∂, L, E, B, G
Finite Element (FEM)	$\partial, \int, L, B, G$
Finite Volume (FVM)	$\int, L, E, B, G$
Spectral methods	F, L, E, B
Discontinuous Galerkin (DG)	$\partial, \int, L, E, B, G$
Boundary Element (BEM)	$\int, L, B, G$
Smoothed Particle (SPH)	∂, N, E, K, G
Lattice Boltzmann (LBM)	E, K, B, G
Density Functional Theory (DFT)	$\partial, L, N, \Pi, B, G, O$
Molecular Dynamics (MD)	N, E, S, K, B
Full Waveform Inversion (FWI)	$\partial, L, E, F, O, B, G$
Tensor Networks (DMRG)	L, Π, O, B
Monte Carlo (MC/MCMC)	N, S, B
Configuration Interaction (CI)	∂, L, Π, B, G
Adaptive Mesh Refinement (AMR)	∂, L, E, B, G
Isogeometric Analysis (IGA)	$\partial, \int, L, B, G$
Radial Basis Functions (RBF)	N, L, B
Peridynamics	$\int, N, E, K, B, G$
Domain Decomposition (DDM)	L, K, B, G
Fluid–Structure Interaction (FSI)	$\partial, L, E, N, K, B, G$
Computed Tomography (CT recon)	$\int, L, O, B, G$
Bayesian Inference (MCMC)	N, S, O, B
Optimal Control	∂, L, E, O, B, G
Compressed Sensing	F, Π, O, B
Particle-in-Cell (PIC)	∂, N, E, S, K, G

We do not claim this basis is provably minimal or sufficient for all possible numerical methods. The 25 families listed are those encountered in our 12 development domains and 72 prospective tasks. Extending $\mathfrak{S}$ to additional primitives is straightforward but requires verification of conditions (2)–(4) for each new primitive.

- other primitive computes ∇^2 .
2. **Integrate** (f): Radiative transfer equation $I(s) = \int_0^s \sigma(s')B(s')e^{-\tau(s,s')}ds'$. The line integral along a ray is irreducible.
 3. **Solve** (L): Stokes flow $-\mu\nabla^2\mathbf{u} + \nabla p = \mathbf{f}$, $\nabla \cdot \mathbf{u} = 0$. Requires solution of a saddle-point linear system.
 4. **Evaluate** (N): Nonlinear Schrödinger equation $i\partial_t\psi = -\nabla^2\psi + |\psi|^2\psi$. The nonlinear term $|\psi|^2\psi$ requires pointwise evaluation.
 5. **Evolve** (E): Heat equation $\partial_t u = \nabla^2 u$. Time integration is irreducible for parabolic evolution.
 6. **Transform** (F): Spectral analysis of turbulence: $E(k) = |\hat{u}(k)|^2$. The energy spectrum requires Fourier transform.
 7. **Project** (Π): Proper Orthogonal Decomposition of flow snapshots. SVD/truncation cannot be expressed via other primitives.
 8. **Sample** (S): Ising model at finite temperature via Metropolis–Hastings. Stochastic sampling is irreducible for thermal averages.
 9. **Couple** (K): Fluid–structure interaction: Navier–Stokes coupled to linear elasticity. The coupling operator is distinct from any single-physics primitive.
 10. **Constrain** (B): Incompressible Navier–Stokes with $\nabla \cdot \mathbf{u} = 0$ enforced via pressure Poisson equation. The divergence-free constraint is irreducible.
 11. **Discretize** (G): Adaptive mesh refinement for a shock tube. The mesh generation/refinement operation cannot be decomposed into other primitives.
 12. **Optimize** (O): Topology optimization (SIMP): $\min_{\rho} \mathbf{f}^T \mathbf{u}$ subject to $\mathbf{K}(\rho)\mathbf{u} = \mathbf{f}$, volume constraint. The optimization loop is irreducible.

S5 Prospective Benchmark: Recruitment and Scientist Details

Table 3. Table S2. Twelve scientists who contributed to the prospective benchmark. None had prior involvement with the framework, the author, or NextGen PlatformAI C Corp.

#	Domain	Institution	Country	Tasks
1	Computational mechanics	MIT	USA	6
2	Quantum chemistry	University of Oxford	UK	6
3	Geophysics	ETH Zürich	Switzerland	6
4	Biomedical imaging	University of Tokyo	Japan	6
5	Atmospheric science	Max Planck Institute	Germany	6
6	Chemical engineering	IIT Bombay	India	6
7	Materials science	Tsinghua University	China	6
8	Astrophysics	University of Cambridge	UK	6
9	Nuclear engineering	CEA Saclay	France	6
10	Ocean acoustics	WHOI	USA	6
11	Plasma physics	Princeton University	USA	6
12	Glacier dynamics	University of Oslo	Norway	6
Total		9 institutions, 8 countries		72

Recruitment protocol. Scientists were identified through three channels: (1) professional networks of the independent coordinator (Dr. M. Torres, ETH Zürich), (2) public calls at SIAM CSE 2025, APS March Meeting 2025, and EGU General Assembly 2025, and (3) direct invitations to authors of recent computational science papers in Nature, Science, and PNAS. Eligibility criteria: active researcher with published computational work, no prior involvement with the framework or the author’s company, and willingness to provide an expert baseline solution with self-reported setup time.

Each scientist received a one-page instruction sheet explaining only how to submit problems (natural language via a web form) and expert baselines (numerical solution files in HDF5 or CSV format). No guidance was provided on problem formatting, difficulty, or domain selection. The $\mathfrak{S}$ -verification procedure was fixed before any problem was executed.

$\mathfrak{S}$ coverage. Of the 72 submitted tasks: 67 were verified as $\in \mathfrak{S}$ by the Judge Agent, 3 were correctly rejected as $\notin \mathfrak{S}$ (genuinely ill-posed), and 2 were rejected due to ambiguous NL input (fixable). The single wrong-answer failure occurred on a problem formally within $\mathfrak{S}$ but near the scientific event horizon (bifurcation point with $L_{\text{DAG}} \gg 1$).

S6 Independent Replication Report

Three researchers independently replicated the pipeline on all 12 development problems:

- **J. Rodriguez** (University of Michigan, PhD candidate in computational mechanics): ran on a 64-core AMD EPYC workstation with 128 GB RAM.
- **A. Patel** (Georgia Institute of Technology, postdoctoral researcher in imaging science): ran on an NVIDIA DGX A100 node.
- **M. Chen** (Stanford University, research scientist in scientific computing): ran on a 32-core Intel Xeon cluster.

Each received the 12 `spec.md` files, the framework codebase (commit hash `c356c318`), and no additional instructions. None had prior involvement with framework development. The Judge Agent verified $\mathfrak{S}$ -membership for all 12 problems on all three hardware configurations.

Table 4. Table S11. Independent replication results across three instances. All problems verified as $\in \mathfrak{S}$; all produced bounded-error solutions.

Problem	Rodriguez	Patel	Chen	Variability
CT reconstruction	31.5 dB	31.8 dB	31.7 dB	± 0.2 dB
Marmousi FWI	26.2 dB	26.5 dB	26.4 dB	± 0.2 dB
GRI-Mech ignition	6.5%	6.1%	6.3%	$\pm 0.2\%$
Granular flow	7.0×10^{-4}	7.4×10^{-4}	7.2×10^{-4}	$\pm 3\%$
He ground state	0.38 mHa	0.42 mHa	0.40 mHa	$\pm 5\%$
BFS turbulent	3.6%	4.0%	3.8%	$\pm 5\%$
Topology opt.	6.0×10^{-4}	6.2×10^{-4}	6.1×10^{-4}	$\pm 2\%$
Waveguide	2.0×10^{-9}	2.2×10^{-9}	2.1×10^{-9}	$\pm 5\%$
Heat equation	3.0×10^{-5}	3.2×10^{-5}	3.1×10^{-5}	$\pm 3\%$
Fresnel diffraction	1.5×10^{-6}	1.7×10^{-6}	1.6×10^{-6}	$\pm 6\%$
Rossby waves	$r = 0.90$	$r = 0.92$	$r = 0.91$	$\pm 1\%$
React.-diffusion	4.8×10^{-4}	5.1×10^{-4}	4.9×10^{-4}	$\pm 3\%$
Bounded error	12/12	12/12	12/12	—

All 12 problems produced bounded-error solutions across all three instances. The variability arises from: (a) stochastic elements in Monte Carlo/Langevin sampling, (b) floating-point non-determinism across hardware, and (c) minor differences in library versions. In all cases, variability is well within the formal error bound guaranteed by Proposition 1.

Table 5. Table S4. Blind challenge: 5 external scientists independently pose research problems. All problems verified as $\in \mathfrak{S}$ by the Judge Agent.

Scientist	Problem Posed	$\mathcal{A}_J$	Bnd.	ρ	Expert Check
Geophysicist	Rayleigh wave in layered soil	$\in \mathfrak{S}$	Yes	$380\times$	Correct
Bioengineer	Drug diffusion through tumor	$\in \mathfrak{S}$	Yes	$520\times$	Correct
Astrophysicist	Bondi accretion onto compact obj.	$\in \mathfrak{S}$	Yes	$450\times$	Correct [†]
Materials sci.	Spinodal decomp. (Cahn–Hilliard)	$\in \mathfrak{S}$	Yes	$600\times$	Correct
Chemical eng.	Packed-bed reactor, mass transfer	$\in \mathfrak{S}$	Yes	$290\times$	Partial [‡]

[†]Matched analytical Bondi profile to 0.3%; missed relativistic corrections (not specified in input). [‡]Mass transfer coefficients matched to 20%; used Sherwood correlation vs. boundary-layer resolution.

Table 6. Table S4b. Framework comparison with domain-specific tools. On any single well-defined PDE, a domain expert using COMSOL or FEniCS will produce a more accurate solution. The pipeline’s contribution is $\mathfrak{S}$ -verified cross-domain automation from natural language, with a median $\rho = 480\times$ setup-time advantage.

Feature	This work	COMSOL	FEniCS	OpenFOAM	Wolfram
Input	NL text	GUI + equations	Python + weak form	Config files	Wolfram Lang.
Domain	$\mathfrak{S}$ verified	Multi-physics	PDE only	CFD only	Broad
Error guarantee	Formal (Prop. 1)	None	A posteriori only	None	None
$\mathfrak{S}$ -verification	Automated	None	None	None	None
ρ (setup time)	$480\times$	—	—	—	—
Expertise req.	Domain only	Domain + numerical	Domain + FEM + Python	Domain + CFD	Domain + Wolfram
Quality audit	Yes ($\mathcal{A}_J$)	Manual	Manual	Manual	Manual

Table 7. Table S5. Five seismic velocity models used for Marmousi FWI validation. All within $\mathfrak{S}$ (regularized inverse problem with bounded L_{DAG}).

#	Model	Complexity	Framework (dB)	Expert (dB)	Ratio
1	Marmousi-2	High (faulted layers)	26.4	27.8	95%
2	Synthetic salt body	Very high (salt dome)	23.1	25.2	92%
3	Thrust fault model	High (overturned layers)	25.4	27.0	94%
4	Thin layers	Medium (thin beds)	27.5	28.6	96%
5	Smooth gradient	Low (1D variation)	28.2	28.9	98%
Mean $\pm$ s.d.			25.8 ± 1.9	27.3 ± 1.6	95%

Table 8. Table S6. GRI-Mech 3.0 ignition delay: per-condition results. Error = $|(\tau_{\text{framework}} - \tau_{\text{exp}})/\tau_{\text{exp}}| \times 100\%$. All conditions within $\mathfrak{S}$; the Judge correctly rejects explicit RK4 at stiffness $> 10^6$ (condition 3 violation) and selects BDF.

T (K)	p (atm)	ϕ	Framework Error (%)	Expert Error (%)
1000	1	1.0	9.2	5.8
1000	10	1.0	8.7	5.1
1000	50	1.0	7.3	4.9
1200	1	1.0	6.8	4.2
1200	10	1.0	6.1	3.5
1200	50	1.0	5.9	3.3
1500	1	1.0	5.5	3.0
1500	10	1.0	5.2	2.8
1500	50	1.0	4.8	2.6
1800	1	1.0	6.4	4.1
1800	10	1.0	5.9	3.4
1800	50	1.0	5.3	3.0
2000	1	1.0	7.8	4.6
2000	10	1.0	7.1	4.0
2000	50	1.0	6.5	3.5
Mean $\pm$ s.d.			6.3 ± 2.1	3.8 ± 1.4

Table 9. Table S7. COVID-19 SEIR-D per-county results. Peak timing error = $|(t_{\text{peak,framework}} - t_{\text{peak,data}})/t_{\text{peak,data}}|$. All within $\mathfrak{S}$; the error is due to model-class limitations (SEIR-D cannot capture intervention timing), not numerical failure.

County	Peak Error (%)	Cum. RMSE (%)	R^2	Expert Peak Err. (%)
King County (WA)	7.3	11.2	0.91	5.1
Maricopa (AZ)	9.8	15.3	0.87	7.2
Fulton County (GA)	11.5	17.8	0.84	8.5
Harris County (TX)	12.1	18.4	0.82	9.0
Los Angeles (CA)	12.8	19.6	0.80	9.4
Cook County (IL)	13.4	20.1	0.79	9.8
Miami-Dade (FL)	14.2	21.8	0.78	10.2
Wayne County (MI)	22.1	25.4	0.71	12.0
Mean $\pm$ s.d.	12.4 ± 3.1	18.7 ± 5.2	0.82 ± 0.09	8.9 ± 2.4

Table 10. Table S8. LLM backbone comparison: Claude-3.5-Sonnet vs. GPT-4 in raw code-generation mode (no Judge, no $\mathfrak{S}$ -verification).

Problem	GPT-4 (raw)		Claude-3.5 (raw)	
	Correct	Bound	Correct	Bound
CT reconstruction	Yes	No	Yes	No
Marmousi FWI	No	No	No	No
GRI-Mech ignition	No	No	Partial	No
Granular flow	No	No	No	No
He ground state	Partial	No	Partial	No
BFS turbulent	No	No	No	No
Topology opt.	Yes	No	Yes	No
Waveguide	Yes	No	Yes	No
Heat equation	Yes	No	Yes	No
Fresnel diff.	Yes	No	Yes	No
Rossby waves	Partial	No	Partial	No
React.-diffusion	Partial	No	Partial	No
Total correct	5/12	0/12	6/12	0/12

Without $\mathfrak{S}$ -verification, neither backbone produces error bounds. Claude-3.5 achieves one additional correct result but still produces no error bounds. This confirms that the Judge Agent’s contribution— $\mathfrak{S}$ -verification—is independent of the underlying LLM.

Table 11. Table S9. Prospective benchmark results stratified by difficulty tier and proximity to $\partial\mathfrak{S}$.

Difficulty	Tasks	Correct+Bounded	Quality Pass	Med. Quality Ratio
Textbook (interior $\mathfrak{S}$)	18	18 (100%)	18 (100%)	98%
Standard (interior $\mathfrak{S}$)	30	27 (90%)	25 (83%)	95%
Frontier (near $\partial\mathfrak{S}$)	24	19 (79%)	15 (63%)	89%
Total	72	64 (89%)	58 (81%)	94%

Difficulty assigned by independent coordinator. Textbook and standard problems lie well within $\mathfrak{S}$ (low L_{DAG}). Frontier problems involve novel coupling, extreme parameters, or high L_{DAG} —closer to the scientific event horizon.

Table 12. Table S10. Per-gate ablation of Judge Agent on 12 development problems. Gates mapped to $\mathfrak{S}$ conditions ($n = 5$ trials each, mean $\pm$ s.d.).

Configuration	Valid	Bounded	$\mathfrak{S}$ condition tested
Full $\mathcal{A}_J$ (all gates + audit)	12/12	12/12	All conditions verified
Remove Gate 1 (dimensional)	11.4/12	10.8/12	Cond. (1): DAG decomposition
Remove Gate 2 (BC/IC)	11.0/12	10.2/12	Cond. (2): well-posedness
Remove Gate 3 (well-posedness)	8.2/12	7.4/12	Cond. (2)+(3): well-posed + stable
Remove Gate 4 (classification)	11.8/12	11.6/12	Cond. (1): correct primitives
Remove Gate 5 (cost estimation)	12/12	11.4/12	Cond. (4): bounded L_{DAG}
Remove quality audit only	12/12	12/12	Post-hoc $\mathfrak{S}$ consistency
Remove all gates (no $\mathcal{A}_J$)	7.4/12	6.8/12	No $\mathfrak{S}$ -verification

The well-posedness gate (conditions 2+3 of $\mathfrak{S}$) is the most critical single gate, catching CFL violations and ill-posed problems that cause 3.8/12 failures when removed. The quality audit is non-redundant: it catches the one qualitative failure (symmetric solution for symmetry-breaking bifurcation near $\partial\mathfrak{S}$) invisible to all pre-execution gates.

Table 13. Table S3. Full list of 50 adversarial test inputs, designed to probe the boundary $\partial\mathfrak{S}$.

#	Category	Problem Description	Outcome
(a) Ambiguous/incomplete NL (20 tests, targeting $\mathfrak{S}$ cond. 1 via $\mathcal{A}_P$)			
1	(a)	“Solve the wave equation” (no BC, no IC, no domain)	Redesign $\rightarrow$ correct
2	(a)	“ $k = 0.1$, solve diffusion” (k ambiguous: conductivity or rate?)	Redesign $\rightarrow$ correct
3	(a)	“Large Reynolds number flow past cylinder” (no Re value)	Redesign $\rightarrow$ correct
4	(a)	“Knudsen regime gas dynamics” (jargon misinterpreted)	Human clarification
5	(a)	“Heat transfer in a room” (no geometry or materials)	Redesign $\rightarrow$ correct
6–10	(a)	Missing domain, missing parameters, ambiguous notation	4 correct, 1 redesign
11–15	(a)	Mixed units, implicit assumptions, domain jargon	5 correct after redesign
16–20	(a)	Contradictory BC, over-specified systems, incomplete models	4 correct, 1 human
(b) Subtle $\mathfrak{S}$-boundary issues (20 tests, targeting conds. 2–4 via $\mathcal{A}_J$)			
21	(b)	Heat eq. with no IC ($\mathcal{I} = \emptyset$) — cond. 2 violation	REJECT (correct)
22	(b)	Nearly incompressible elasticity ($\nu = 0.4999$) — near $\partial\mathfrak{S}$	Accept (locking [†])
23	(b)	Reaction-diffusion with extreme ratio (10^{12}) — cond. 4	Accept (caught by $\mathcal{A}_E$)
24	(b)	Predict which slit a photon passes through — $\notin \mathfrak{S}$	REJECT (correct)
25	(b)	Near-singular Helmholtz at resonance — cond. 2	REJECT (correct)
26–30	(b)	Ill-conditioned systems, degenerate BCs — conds. 2–3	4 reject, 1 accept
31–35	(b)	Near-critical parameters, bifurcation — cond. 4 ($\partial\mathfrak{S}$)	3 reject, 2 correct
36–40	(b)	Stiff systems near stability boundary, chaotic regimes	5 correct
(c) Event-horizon probes (10 tests, L^2 correct but qualitatively wrong)			
41	(c)	Diffusion with non-monotone numerical solution	Audit catches
42	(c)	Symmetry-breaking bifurcation (symmetric L^2 minimizer)	Audit misses [‡]
43	(c)	Shock without entropy condition (smooth approximation)	Audit catches
44	(c)	Reaction front with negative species concentrations	Audit catches
45	(c)	Bifurcation near critical Re (wrong branch) — $\partial\mathfrak{S}$	Audit misses [‡]
46–50	(c)	Conservation violations, wrong attractor, missing boundary layer	4 audit catches, 1 miss

[†] $\mathcal{A}_J$ false negative: condition number $> 10^{12}$ but coercivity formally satisfied; near $\partial\mathfrak{S}$ (cond. 4). [‡]Both misses are at the scientific event horizon where $L_{\text{DAG}} \rightarrow \infty$; bifurcation detection remains an open problem (see proposed heuristics in main text).

Additional Tables

S7 LLM Backbone Reproducibility and Version Sensitivity

The pipeline currently uses Claude-3.5-Sonnet (Anthropic, version 20241022) as the backbone for all three agents. To ensure reproducibility and assess sensitivity to model version:

Cached inference logs. All LLM API calls during the 12 development problems, 72 prospective tasks, and 50 adversarial tests are recorded with full request/response payloads. These logs enable complete reproduction of results without API access, and are included in the Zenodo archive.

Multi-backbone testing. We re-ran the 12 development problems with two alternative LLM backbones serving as $\mathcal{A}_P$ and $\mathcal{A}_E$ (the $\mathcal{A}_J$ $\mathfrak{S}$ -verification logic is deterministic and backbone-independent):

Table 14. Table S14. LLM backbone sensitivity on 12 development problems. $\mathfrak{S}$ -verification is backbone-independent.

Backbone	Correct	Bounded	Median Quality Ratio
Claude-3.5-Sonnet (default)	12/12	12/12	94%
Claude-3-Opus	12/12	12/12	93%
GPT-4-Turbo	11/12	11/12	91%

GPT-4-Turbo failure: GRI-Mech ignition delay. $\mathcal{A}_P$ generated a spec with explicit Euler (stiffness ratio 10^{10}); $\mathcal{A}_J$ correctly rejected ($\mathfrak{S}$ condition 3 violated) and triggered redesign, but GPT-4-Turbo’s redesign selected LSODA with insufficient absolute tolerance, producing 18% error (above the 10% threshold). The $\mathfrak{S}$ -verification caught the initial violation; the redesign failure is an LLM capability issue, not a $\mathfrak{S}$ -verification issue.

Fragility disclosure. We cannot guarantee that future LLM updates will preserve current performance. The $\mathcal{A}_J$ $\mathfrak{S}$ -verification gates provide a safety net—they will reject specifications that violate $\mathfrak{S}$ conditions regardless of which LLM generated them—but the pipeline’s *success rate* depends on $\mathcal{A}_P$ ’s ability to generate correct specifications within 3 redesign attempts. We recommend pinning model versions for production use and re-validating after any model update.

Table 15. Table S12. Extended controlled comparison on 72 prospective tasks. Without $\mathfrak{S}$ -verification (No $\mathcal{A}_J$), success drops to 53%.

Difficulty tier	Full pipeline	No $\mathcal{A}_J$	GPT-4+spec	PINN
Textbook (interior $\mathfrak{S}$, 18)	18/18 (100%)	14/18 (78%)	13/18 (72%)	11/18 (61%)
Standard (interior $\mathfrak{S}$, 30)	27/30 (90%)	17/30 (57%)	13/30 (43%)	7/30 (23%)
Frontier (near $\partial\mathfrak{S}$, 24)	19/24 (79%)	7/24 (29%)	5/24 (21%)	1/24 (4%)
Total	64/72 (89%)	38/72 (53%)	31/72 (43%)	19/72 (26%)

The Judge’s value increases monotonically with proximity to $\partial\mathfrak{S}$: 22 percentage-point gap on textbook problems vs. 50 percentage-point gap on frontier problems.

Supplementary Figures

Figure S1. CT per-case PSNR distribution. Histogram of PSNR across 200 LoDoPaB-CT test cases for the framework (blue) and expert baseline (orange). The framework distribution is slightly left-shifted (mean 31.7 vs. 32.1 dB) with comparable spread ($\sigma = 1.2$ vs. 1.1 dB). The 95% CI for the mean difference is $[-0.6, -0.2]$ dB (bootstrap, $n = 10,000$). Cohen’s $d = 0.35$. CT reconstruction lies well within $\mathfrak{S}$ (low $L_{\text{DAG}} \approx 2.1$).

Table 16. Table S13. Per-gate ablation of Judge Agent on 72 prospective tasks. Gates mapped to $\mathfrak{S}$ conditions ($n = 3$ trials each).

Configuration	Correct + Bounded	Quality Pass	$\mathfrak{S}$ cond.
Full $\mathcal{A}_J$ (all gates + audit)	64/72 (89%)	58/72 (81%)	All
Remove well-posedness gate	50/72 (69%)	44/72 (61%)	2+3
Remove BC/IC compatibility	56/72 (78%)	48/72 (67%)	2
Remove dimensional consistency	60/72 (83%)	52/72 (72%)	1
Remove classification	62/72 (86%)	56/72 (78%)	1
Remove cost estimation	64/72 (89%)	54/72 (75%)	4
Remove quality audit only	64/72 (89%)	52/72 (72%)	Post-hoc
Remove all (no $\mathcal{A}_J$)	38/72 (53%)	28/72 (39%)	None

The well-posedness gate ($\mathfrak{S}$ conditions 2+3) remains the most critical single gate. The quality audit’s contribution is most visible in the gap between “Correct + Bounded” and “Quality Pass”: removing the audit does not change the bounded-error count but allows 6 additional qualitative failures (near $\partial\mathfrak{S}$) to pass undetected.

Figure S2. Prospective benchmark quality ratio distribution. Boxplot of quality ratio (framework / expert baseline) for the 58 fully successful prospective tasks. Median: 94%. IQR: 89–98%. Whiskers: 78–100%. One outlier at 72% (a highly optimized turbulence simulation where the expert used a custom adaptive scheme—a frontier problem near $\partial\mathfrak{S}$). Stratified by domain: imaging tasks (median 97%), PDE tasks (93%), optimization tasks (91%), stochastic tasks (88%).

Figure S3. Scalability, ρ , and $\mathfrak{S}$ -verification overhead. Left panel: wall-clock time breakdown ($\mathcal{A}_P$, $\mathcal{A}_J$ $\mathfrak{S}$ -verification, $\mathcal{A}_E$) across the 12 development problems, ordered by total DOF. $\mathcal{A}_J$ overhead: 3% (heat equation, 10^3 DOF) to 28% (GRI-Mech, stiffness analysis dominates). $\mathcal{A}_E$ dominates above 10^4 DOF. Right panel: $\mathcal{A}_J$ time vs. total time as a function of problem DOF; the ratio decreases as $O(1/\text{DOF})$ since $\mathfrak{S}$ -verification cost is nearly constant while $\mathcal{A}_E$ scales with problem size. The setup-time efficiency ρ is independent of $\mathcal{A}_J$ overhead, since ρ measures total setup time (minutes) vs. expert setup time (days).